A split admitting a rank and degree extension

Human Compression project

A perfect split is internal when its label is neither endpoint label. Its crossing mark has the form

$$q = (a\ b)\alpha\beta,$$

where the prefix and suffix are supported and transitive on disjoint sets A, B, the crossing points belong to their respective sets, and the internal factors are supported off a, b on the corresponding sides. Call the split *sufficient* if q is pure, or if one internal factor is trivial and both actual attaching groups contain an odd permutation. When $\alpha = 1$ these groups are the prefix L and the group J generated by q and the suffix. Their supports meet at a. This definition retains the exact hypotheses of the acquired pure-split and odd-attachment theorems.

The acquired components include the complete parity proof for the remaining middle, its exact finite boundaries and their distinct reviews. We now close the induction on the actual degree.

Theorem 1. *Let a fractured string C-marking of the natural symmetric group S_n, with $n \geq 5$, have rank r satisfying $2r > n + 2$. It has an internal sufficient split. For $n \geq 7$, the fracture hypothesis follows from the same rank inequality and need not be imposed separately.*

Proof. We use strong induction on the actual degree. The acquired orbit forest gives $r \leq n - 1$, so degrees at most four cannot meet the strict inequality. In degrees five and six, that inequality forces rank $n - 1$. The acquired complete extremal classification gives the ordinary adjacent-transposition simplex, apart from the outer degree-six action. The latter has a transitive maximal omission and is excluded by fracture. Every internal simplex mark is a pure perfect split. These are the complete induction bases.

Suppose $n \geq 7$ and the statement is known at every smaller degree. Apply the acquired conditional reduction to even anchored factors. Its complete hypotheses are exactly the present natural symmetric marking, fracture, strict rank inequality and smaller-degree sufficient-split induction. It uses the already proved trivial-factor theorem, actual-wing theorem, even-attachment bound and one-point transport, so no split-existence or quotient-intersection assumption is being added here. Its conclusion is a sufficient split, or the following actual residual, after reversal:

factor	degree	rank
P, natural symmetric	m even	$m/2 + 1$
E, even	$e \in \{6, 8, 10\}$	$e/2$.

The label intervals are consecutive and disjoint; both factors are fractured and transitive on supports meeting only at their common endpoint anchor. Their full degree and rank satisfy $n = m + e - 1$ and $2r = n + 3$.

The acquired small-even theorem applies to E. Its external group is the actual odd transitive group P on the other support, meeting it only at the anchor, within the original C-marking. These are precisely its attachment and supported-kernel hypotheses. Thus $e = 6$ and E is the primitive degree-six rank-three A_5 factor. The complete first-cut and endpoint arguments, followed by the paired-middle parity theorem, supply a sufficient split in exactly this critical configuration. Their uses of induction are exclusively in actual symmetric factors of degree strictly less than the present n, with the same strict inequality. Their separate rank induction

for paired middles uses only anchor deletion and finite boundaries, not the theorem now being proved. This completes strong induction.

Finally the acquired initial-fracture theorem applies to every natural symmetric marking with $n \geq 7$ and $2r > n + 2$, with its complete exceptional-rank inventory retained. It supplies the last assertion without a new finite premise. □

Corollary 2. *For every natural symmetric string C-marking of degree $n \geq 5$ and rank r with $2r > n+2$, there is a natural symmetric string C-marking of degree $n+1$ and rank $r+1$ obtained by subdividing an internal perfect crossing. At degree six the marking may first be replaced by an abstractly isomorphic natural representative.*

Proof. For $n \geq 7$, Theorem 1 supplies the split. If it is pure, use the acquired pure-split theorem. Otherwise reverse labels if needed to make $\alpha = 1$. At a new point c, replace

$$q = (a\ b)\beta \quad \text{by} \quad (a\ c),\ (c\ b)\beta.$$

The acquired odd-attachment theorem verifies both complete leaf kernel families from the original IP and the two odd attaching groups. Its exact split criterion then gives the full intersection property and string relations. The new group contains the old S_n fixing c, since conjugating the second new mark by the first recovers q. It also contains the pure transposition $(a\ c)$, so it is S_{n+1}. Rank and degree both increase by one.

At degrees five and six the full extremal classification again reduces the abstract marking to the simplex, whose subdivision is the next simplex. The outer action of S_6 is abstractly isomorphic to that marking, though it was correctly excluded from the fractured natural-action theorem. This accounts for the distinction between actual permutation actions and abstract string C-markings. □

Attribution and unproved counting step. The source is Cameron–Fernandes–Leemans, *The number of string C-groups of high rank*, arXiv:2212.12723v2. The actual subdivision is their rank and degree operation. All rank classifications and finite lists remain attributed in the acquired components. The present proof combines the repaired restriction criterion, odd-attachment lemma and the completed critical route. It claims existence of a valid extension. It does not yet prove an injection or bijection on markings modulo isomorphism and duality: an intrinsic choice, a valid contraction, and compatibility with reversal remain to be established. No whole-original credit, new conjecture, Lean theorem or compression measurement is assigned.